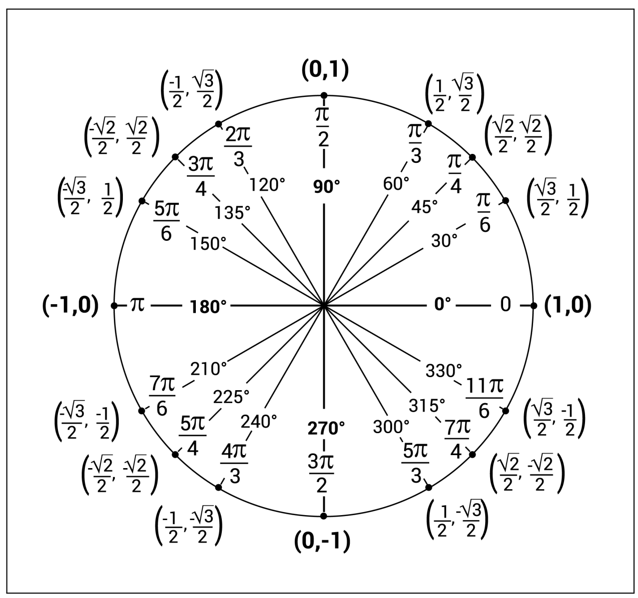
Math 125

Extra Unit A practice 3

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the past final worksheets, and the Brightspace problems. The formulas below will be provided on Opportunity A and Jubilee 1.

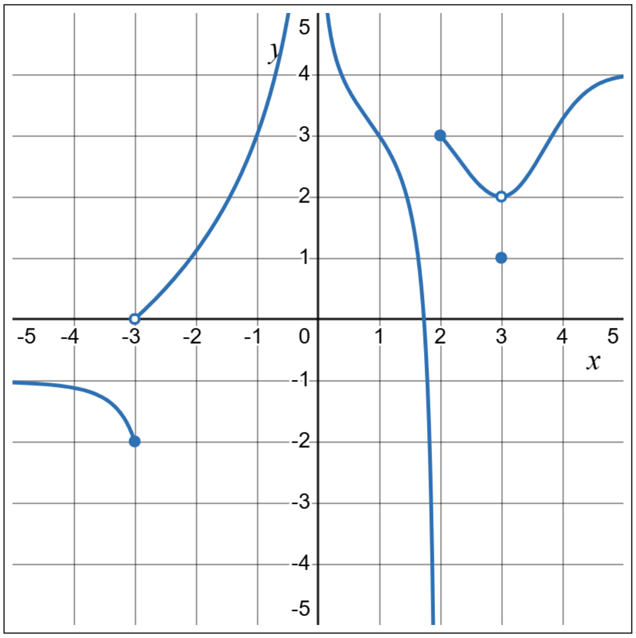
Formulas

- $\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1$, $\lim_{\theta \to 0} \frac{\cos\theta - 1}{\theta} = 0$
- Continuity at $x = a$: $\lim_{x \to a} f(x) = f(a)$
- $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$
- Basic derivatives
  - $\frac{d}{dx}(c \cdot f(x)) = c \cdot f'(x)$
  - $\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$
  - $\frac{d}{dx}(x^n) = nx^{n-1}$
  - $\frac{d}{dx}(e^x) = e^x$
  - $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$
  - $\frac{d}{dx}(\sin x) = \cos x$
  - $\frac{d}{dx}(\cos x) = -\sin x$
  - $\frac{d}{dx}(\tan x) = \sec^2 x$
  - $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$
  - $\frac{d}{dx}(\cot x) = -\csc^2 x$
  - $\frac{d}{dx}(\csc x) = -\csc x \cdot \cot x$
- Derivative rules
  - $\frac{d}{dx}(f(x) \cdot g(x)) = f'(x) \cdot g(x) + g'(x) \cdot f(x)$
  - $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
  - $\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$
- $L(x) = f'(a)(x-a) + f(a)$



A1 - Limits and continuity

(a) The graph of f is given. Use the graph to evaluate each limit if it exist. If it does not, write DNE.



(i) $\lim_{x\to 2^+} f(x)$ $\quad 3$

(ii) $\lim_{x\to -3} f(x)$ $\quad$ DNE

(iii) $\lim_{x\to 0} f(x)$ $\quad \infty$

(iv) $\lim_{x\to -\infty} f(x)$ $\quad -1$

(b) Find the limits. Clearly show each step of your solution for full credit. If the limit does not exist, explain why.

(i) $$\lim_{x\to\infty} \frac{\sqrt{x^2+4x+1}-x}{1} \cdot \frac{\sqrt{x^2+4x+1}+x}{\sqrt{x^2+4x+1}+x} = \lim_{x\to\infty} \frac{4x+1}{\sqrt{x^2+4x+1}+x} \cdot \frac{\frac{1}{x}}{\frac{1}{x}} = \lim_{x\to\infty} \frac{4+\cancel{\frac{1}{x}}^{\,0}}{\sqrt{1+\cancel{\frac{4}{x}}^{\,0}+\cancel{\frac{1}{x^2}}^{\,0}}+1} = \frac{4}{\sqrt{1}+1} = \boxed{2}$$

(ii) $$\lim_{v\to 4^+} \frac{4-v}{|4-v|} = \lim_{v\to 4^+} \frac{4-v}{-(4-v)} = \boxed{-1}$$

$4-v<0$

when $v>4$

(c) Consider the function f that is given by

$$f(x) = \begin{cases} x+1 & x<1 \\ c & x=1 \\ 5x^2-4 & x>1 \end{cases}$$

For what value(s) of c is the function continuous at $x=1$? Write "none" if there are none. Explain.

$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} x+1 = 2$

$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} 5x^2-4 = 1$

$\neq$ so no such c exists.

A2 - The derivative

Consider the functions $f(x) = \sqrt{2x+6}$ and $g(x) = \dfrac{32}{x+3}$.

(a) Use the limit definition of the derivative to find $f'(-3)$ or show that it doesn't exist. No credit will be awarded if you don't use the limit definition of the derivative.

$$f'(-3) = \lim_{h\to 0} \frac{\sqrt{2(-3+h)+6} - \sqrt{2\cdot -3+6}}{h} = \lim_{h\to 0} \frac{\sqrt{2h}}{h} = \lim_{h\to 0} \frac{\sqrt{2}}{\sqrt{h}} = \infty$$

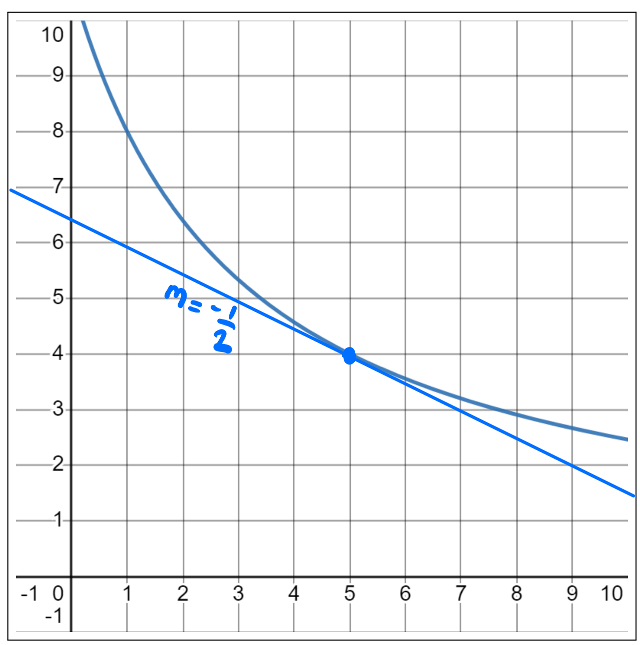
$\rightarrow$ $f'(-3)$ does not exist

(b) Use the limit definition of the derivative to find $g'(5)$ or show that it doesn't exist. No credit will be awarded if you don't use the limit definition of the derivative.

$$g'(5) = \lim_{h\to 0} \frac{\frac{32}{5+h+3} - \frac{32}{5+3}}{h} = \lim_{h\to 0} \frac{\frac{32}{8+h} - 4}{h} \cdot \frac{8+h}{8+h} = \lim_{h\to 0} \frac{32-32-4h}{h(8+h)}$$

$$= \lim_{h\to 0} \frac{-4}{8+h} = \boxed{-\frac{1}{2}}$$

(c) A graph of $g(x)$ is shown to the right. Sketch a graphical representation of the value $g'(5)$. You should clearly indicate what about your sketch represents the value $g'(5)$. Note that you can get this question correct even if you can't do part (b).



A3 - Derivative rules

Find the derivative of each function. You do not need to simplify your answer.

(a) $g(x) = (-6x-9)^{\sqrt{7x^2+3x-1}}$

$$g(x) = e^{(7x^2+3x-1)^{\frac{1}{2}} \cdot \ln(-6x-9)}$$

$$g'(x) = e^{(7x^2+3x-1)^{\frac{1}{2}} \cdot \ln(-6x-9)} \cdot \left(\frac{1}{2}(7x^2+3x-1)^{-\frac{1}{2}} \cdot (14x+3) \cdot \ln(-6x-9) + (7x^2+3x-1)^{\frac{1}{2}} \cdot \frac{-6}{-6x-9} \right)$$

(b) $f(x) = \dfrac{e^{\tan(x)}}{x}$

$$f'(x) = \frac{x \cdot e^{\tan x} \cdot \sec^2 x - e^{\tan x}}{x^2}$$

A4 - Implicit curves and linearization

(a) Consider the curve given by the equation

$$y = \ln(x^2 - \sin(y) - 8)$$

Find the equation of the tangent line to this curve at the point $(3, 0)$.

$\frac{dy}{dx} = \frac{2x - \cos y \cdot \frac{dy}{dx}}{x^2 - \sin(y) - 8}$ @ $(3,0) \to$ $\frac{dy}{dx} = \frac{2 \cdot 3 - \cos(0) \cdot \frac{dy}{dx}}{3^2 - \sin(0) - 8} = 6 - \frac{dy}{dx}$

$\to \frac{dy}{dx} = 3$

$\boxed{y = 3(x-3) + 0}$

(b) Use a linear approximation to estimate $(8.06)^{2/3}$.

$f(x) = x^{\frac{2}{3}}$ $\qquad f(8) = 4$

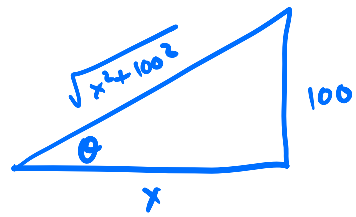
$f'(x) = \frac{2}{3} x^{-\frac{1}{3}}$ $\qquad f'(8) = \frac{1}{3}$

$L(x) = \frac{1}{3}(x-8) + 4$

$f(8.06) \approx L(8.06) = \frac{1}{3} \cdot (8.06 - 8) + 4 = \boxed{4.02}$

A5 - Related rates

(a) How fast is the length of the shadow cast by a 100-foot building changing when the shadow is 20 feet long, assuming the angle of elevation to the sun is decreasing at 0.25 radians per hour? Include units in your answer.



$$\frac{(20^2+100^2)\cdot -0.25}{-100} = \frac{dx}{dt}$$

$$\tan\theta = \frac{100}{x}$$

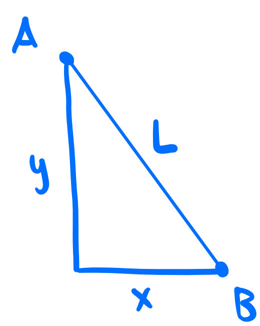
$$\sec^2\theta \cdot \frac{d\theta}{dt} = -100\, x^{-2} \cdot \frac{dx}{dt}$$

$$\frac{x^2+100^2}{\cancel{x^2}} \cdot \frac{d\theta}{dt} = \frac{-100 \cdot \frac{dx}{dt}}{\cancel{x^2}}$$

26 ft/hr

(b) At noon, boat A is 17 miles west of boat B. Boat A is sailing directly north at 5 miles per hour. Boat B is sailing in such a way that its westward speed is 1 mile per hour and its northward speed is 1 mile per hour. How fast is the distance between the boats changing two hours later at 2pm? Include units in your answer.

Hint: If you draw a right triangle representing the position of the boats at 2pm, the length of the vertical leg of the right triangle is changing at a rate equal to the difference of the northward speeds of the two boats.



$$\frac{dy}{dt} = 5 - 1 = 4 \text{ mph}$$

$$\frac{dx}{dt} = -1 \text{ mph}$$

@ 2pm

$y = 8$

$x = 15$

$\rightarrow L = \sqrt{15^2+8^2} = 17$

$$x^2 + y^2 = L^2$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 2L \cdot \frac{dL}{dt}$$

$$\frac{dL}{dt} = \frac{x \cdot \frac{dx}{dt} + y \cdot \frac{dy}{dt}}{L} = \frac{15 \cdot -1 + 8 \cdot 4}{17} = 1$$

1 mph